Valuation of Options - part 1

Quantitative Finance

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Where this fits: the course at a glance

Options recap → Part 1 binomial → Part 2 PDE → Part 3 risk-neutral → Hedging (B) → MC Greeks (C)

One question: how to *price* and *hedge* a European option — and why it works.

Three ideas recur throughout:

- *three methods, one price*: replication = risk-neutral = pricing kernel;
- *discrete* $\leftrightarrow$ *continuous*: binomial $q \leftrightarrow$ market price of risk; hedge ratio $\leftrightarrow$ delta; tree $\leftrightarrow$ finite-difference grid;
- *error* $\sim 1/\sqrt{n}$: discrete-time hedging and Monte-Carlo estimation alike.

Section 1

Introduction

Options (prerequisites)

We assume familiarity with the basics of options — calls and puts, payoffs at maturity, payoff diagrams, moneyness, and exercise styles. These are collected in the companion deck *"Options: a quick recap"*.

Throughout, S_t denotes the price of the underlying at time t, K the strike, and T the maturity; a European payoff is written $C_T = g(S_T)$.

Main problems

Unlike a stock, an option has no obvious "correct" price: its value is *derived* from the underlying. This raises two central problems.

1. Valuation/pricing. What is the 'fair' price C_t of an option, and how does it evolve over time? This matters for:

- investors/traders — to avoid overpaying, or to profit from a market price they believe is wrong;
- financial institutions — to quote prices for OTC (or 'embedded') options, and to value and risk-manage books of options.

2. Hedging. Having sold an option, how do you control the resulting risk? (Treated later in the course.)

A striking feature of the models we use: these two problems turn out to share the *same* solution.

- we are interested in (fair) price C_t, for $t \in [0, T)$, of (European) option with payoff $C_T = g(S_T)$ at maturity T
- we will learn three methods to determine 'fair price' (using specified continuous-time model for financial market):
    - Black-Scholes Partial Differential Equation
    - risk-neutral pricing
    - pricing kernel
- in particular: you will derive the famous Black-Scholes price for European put option with maturity T and strike K:

$$C_t = p(S_t, T - t, K, r, \sigma) = K\,\mathrm{e}^{-r(T-t)}\,\Phi(-d_2) - S_t\Phi(-d_1)$$

$$d_1 = \frac{\log\left(\frac{S_t}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t},$$

    - does not depend on μ!

Black-Scholes formula

- in 1973 Fisher Black and Myron Scholes published paper 'The Pricing of Options and Corporate Liabilities' and Robert Merton published 'Theory of Rational Option Pricing'
- Merton and Scholes received Nobel Memorial Prize in Economic Sciences in 1997 (Black died in 1995)

Arbitrage

All pricing methods we discuss rely on insisting that arbitrage opportunities do/should not exist.

Definition

If it is possible to find an admissible self-financing trading strategy with value process V such that, for some $T > 0$, we have $V_0 \leq 0$ and

$$\mathbb{P}(V_T < 0) = 0 \text{ and } \mathbb{P}(V_T > 0) > 0,$$

then we say that an arbitrage opportunity exists.

Remarks:

- in QF you are allowed to ignore 'admissible': this condition rules out 'doubling strategies'. A sufficient condition for admissibility: $\mathbb{E}V_t^2 < \infty$ for all $t \in [0, T]$, where V_t denotes the portfolio value of the self-financing strategy

A first no-arbitrage result: put-call parity

No arbitrage already pins down some prices *without any model*. Take a European call and put with the same strike K and maturity T.

- at T: $(S_T - K)^+ - (K - S_T)^+ = S_T - K$, so *long call – short put* pays exactly $S_T - K$;
- but *one share* $-$ K *zero-coupon bonds* pays the same $S_T - K$ at T;
- equal payoffs $\Rightarrow$ equal price today (law of one price):

$$\boxed{C_t - P_t = S_t - K\, e^{-r(T-t)}} \qquad \text{(put-call parity)}.$$

Forward price. The same reasoning gives the no-arbitrage price for delivery of the stock at T:

$$F_t = S_t\, e^{r(T-t)}.$$

Neither relation uses the Black-Scholes model — only no arbitrage.

No-arbitrage bounds on the call price

Even before we price it, no arbitrage bounds a European call:

$$\left(S_t - K\,\mathrm{e}^{-r(T-t)}\right)^+ \;\le\; C_t \;\le\; S_t.$$

- *upper bound* $C_t \le S_t$: the call never pays more than the stock, so it cannot cost more (else short the call, buy the stock);
- *lower bound*: from parity $C_t = P_t + S_t - K\,\mathrm{e}^{-r(T-t)} \ge S_t - K\,\mathrm{e}^{-r(T-t)}$ (as $P_t \ge 0$), and $C_t \ge 0$;
- since $S_t - K\,\mathrm{e}^{-r(T-t)} > S_t - K$, the call is worth *more* than its intrinsic value $(S_t - K)^+$ — so on a non-dividend stock an **American call is never exercised early**.

Section 2

Toy model

Motivation

First we discuss a very simple discrete time model and discuss pricing of options under assumption that arbitrage opportunities do not exist. We discuss three methods:

- pricing by replication
- risk-neutral pricing
- the pricing kernel

Later on we will discuss their continuous-time analogues. Although the math gets more complicated, the underlying ideas are the same!

Model for financial market

- two states-of-the-world: $\Omega = \{\text{“up”}, \text{“down”}\}$
- two assets: risky asset S and riskless asset B
  - $B_1(\text{“up”}) = B_1(\text{“down”}) = B_1 > 0$
  - for

$$S_1(\omega) = \begin{cases} s_u, & \text{if } \omega = \text{“up”} \\ s_d, & \text{if } \omega = \text{“down”} \end{cases}$$

    we assume $0 < s_d < s_u$
- probability of state “up” is $p \in (0, 1)$, i.e. $\mathbb{P}(\text{“up”}) = p$

$$\begin{tikzcd}
 & \begin{pmatrix} s_u \\ B_1 \end{pmatrix} \\
\begin{pmatrix} S_0 \\ B_0 \end{pmatrix} \arrow[ur, "p"] \arrow[dr, "1-p"'] & \\
 & \begin{pmatrix} s_d \\ B_1 \end{pmatrix}
\end{tikzcd}$$

No arbitrage

Theorem There are no arbitrage opportunities in our financial market if and only if

$$0 < \frac{s_d}{S_0} < \frac{B_1}{B_0} < \frac{s_u}{S_0}.$$

Proof
'$\Rightarrow$': exercise
'$\Leftarrow$': this follows from the risk-neutral construction derived later in this section (verify!)

A full proof of '$\Rightarrow$' is given in the appendix.

What's the price of an option in this market?

- now consider European option with payoff

$$C_1(\omega) = \begin{cases} c_u, & \text{if } \omega = \text{“up”} \\ c_d, & \text{if } \omega = \text{“down”} \end{cases}$$

  - for call: $c_u = \max\{s_u - K, 0\}$ and $c_d = \max\{s_d - K, 0\}$
- what is price at $t = 0$ for call?

$$\begin{tikzcd} & \begin{pmatrix} s_u \\ B_1 \end{pmatrix} & C_1 = c_u \\ \begin{pmatrix} S_0 \\ B_0 \end{pmatrix} \arrow[ur, "p"] \arrow[dr, "1-p"'] & & \\ & \begin{pmatrix} s_d \\ B_1 \end{pmatrix} & C_1 = c_d \end{tikzcd}$$

Method 1: pricing by replication

- construct portfolio V with $V_1(\omega) = C_1(\omega)$ for all ω
  - as portfolio generates exactly the same pay-offs as option does (for all states-of-the-world), the portfolio is called *replicating* portfolio
  - Why? As there are no intermediate cashflows, we must have $C_0 = \phi S_0 + \psi B_0$. Otherwise we would create arbitrage opportunities (this is the *law of one price*). So via the replicating portfolio we find the no-arbitrage price of the option
- notice that we have to determine ϕ and ψ such that

$$c_u = \phi s_u + \psi B_1$$
$$c_d = \phi s_d + \psi B_1$$

which yields

$$\phi = \frac{c_u - c_d}{s_u - s_d} \text{ and } \psi = \frac{c_d s_u - s_d c_u}{B_1(s_u - s_d)}$$

- price of option (at $t = 0$):

$$C_0 = \phi S_0 + \psi B_0$$

Method 2: risk-neutral pricing

Motivation

- people would like to have pricing formula of the form:

$$\frac{A_0}{B_0} = \mathbb{E}\left[\frac{A_1}{B_1}\right]$$

  for all assets A, i.e. price equals expected discounted cashflows
- note that

$$\frac{A_0}{B_0} = \mathbb{E}\left[\frac{A_1}{B_1}\right] \quad \Longleftrightarrow \quad \mathbb{E}\left[\frac{A_1}{A_0}\right] = \frac{B_1}{B_0}$$

- however, already for $A = S$ this almost never holds true in real financial markets:

$$\mathbb{E}\left[\frac{S_1}{S_0}\right] = \frac{B_1}{B_0},$$

  would mean that the expected (gross) return on the stock equals the riskless return — i.e. that investors demand *no risk premium* for bearing risk, which is not what we observe

Intermezzo: a biased coin can save the day

- Suppose we throw a *fair* coin, $\Omega = \{H, T\}$, and you get pay-off $X = 1$ if $\omega = H$ and $X = 0$ if $\omega = T$
- the price of entering this game is 0.3 (given)
- (for some reason) we want to have the identity

$$\text{price} = \mathbb{E}[X] = \text{Prob}(\{H\})$$

- as the coin is fair the equation does not hold true if we use the true, "real-world", probability measure ($0.3 \neq 0.5$)
- however: we can just create *artificial* probability measure $\mathbb{Q}$ with $\mathbb{Q}(\{H\}) = 0.3$, which does lead to

$$\text{price} = \mathbb{E}_{\mathbb{Q}}[X]$$

- back in the market, the role of the given "price 0.3" is played by today's asset prices: we will choose $\mathbb{Q}$ so that it reproduces S_0 and B_0.

Method 2: risk-neutral pricing

- people would like to have pricing formula of the form:

$$\frac{A_0}{B_0} = \mathbb{E}\left[\frac{A_1}{B_1}\right] \iff \mathbb{E}\left[\frac{A_1}{A_0}\right] = \frac{B_1}{B_0} \quad (\star)$$

  for all assets A
- if we use "real-world" probabilities then this identity will typically not hold for $A = S$
- could we repair identity by using artificial probability measure $\mathbb{Q}$ instead of "real-world" measure $\mathbb{P}$?
- solving

$$\frac{S_0}{B_0} = \mathbb{E}_{\mathbb{Q}}\left[\frac{S_1}{B_1}\right] = q\frac{s_u}{B_1} + (1-q)\frac{s_d}{B_1}$$

  yields

$$q = \frac{\frac{B_1}{B_0} - \frac{s_d}{S_0}}{\frac{s_u}{S_0} - \frac{s_d}{S_0}}$$

- if $q \in [0,1]$ then we indeed have found artificial probability measure for which $(\star)$ is true by construction (for $A = B$ the identity is trivial)

Method 2: risk-neutral pricing

Recall earlier theorem: there is no arbitrage if and only if

$$0 < \frac{s_d}{S_0} < \frac{B_1}{B_0} < \frac{s_u}{S_0}$$

which occurs exactly if $q \in (0, 1)$. So we have proved:

Intermediate result
The market is free of arbitrage opportunities if and only if there exists a probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ such that S/B is a martingale under $\mathbb{Q}$, i.e.

$$\frac{S_0}{B_0} = \mathbb{E}_{\mathbb{Q}}[\frac{S_1}{B_1}].$$

Method 2: risk-neutral pricing

Terminology:

- As S/B is martingale under $\mathbb{Q}$ we have

$$\frac{S_0}{B_0} = \mathbb{E}_{\mathbb{Q}}\left[\frac{S_1}{B_1}\right],$$

  which yields

$$\frac{B_1}{B_0} = \mathbb{E}_{\mathbb{Q}}\left[\frac{S_1}{S_0}\right],$$

  i.e. the expected (gross) return on the risky asset S, under $\mathbb{Q}$, is equal to the return/interest on the riskless asset B.
- $\mathbb{Q}$ is often called "risk-neutral measure"
  - Please note that $\mathbb{Q}$ is *artificial* measure and does not describe actual probabilities!
- $\mathbb{Q}$ is also called *equivalent martingale measure*

Method 2: risk-neutral pricing

So we have repaired the formula $S_0/B_0 = \mathbb{E}[S_1/B_1]$, but it was quite a discussion. What is the true benefit?

- introduce, as before, an option to the market (with payoff C_1 at $t = 1$)
- we already know that there is a replicating portfolio (ϕ, ψ) such that $V_1(\omega) = \phi S_1(\omega) + \psi B_1 = C_1(\omega)$ for all ω
- and, if there are no arbitrage opportunities, $C_0 = \phi S_0 + \psi B_0$
- now note that

$$\mathbb{E}_{\mathbb{Q}}\left[\frac{C_1}{B_1}\right] = \phi \mathbb{E}_{\mathbb{Q}}\left[\frac{S_1}{B_1}\right] + \psi = \phi \frac{S_0}{B_0} + \psi = \frac{1}{B_0}(\phi S_0 + \psi B_0) = \frac{C_0}{B_0},$$

  so we can calculate for any option (or asset) its price at $t = 0$ via the above identity as soon as we know $\mathbb{Q}$! There is no need to calculate the replicating portfolios.

Worked example: one-period model

Setup (numbers from the accompanying spreadsheet): $B_0 = 1$, $B_1 = 1.02$ (so $r = 2\%$), real-world probability $p = \mathbb{P}(\text{up}) = 0.7$, and the claim is a call with strike $K = 50$.

$$\begin{tikzcd}
 & S_{1,u} = 150, \quad c_u = 100 \\
S_0 = 125, \quad B_0 = 1 \arrow[ur, "p = 0.7"] \arrow[dr, "1 - p = 0.3"'] & \\
 & S_{1,d} = 100, \quad c_d = 50
\end{tikzcd}$$

Worked example: pricing by replication

Solve $c_u = \phi s_u + \psi B_1$ and $c_d = \phi s_d + \psi B_1$:

$$\phi = \frac{c_u - c_d}{s_u - s_d} = \frac{100 - 50}{150 - 100} = 1,$$

$$\psi = \frac{c_d s_u - s_d c_u}{B_1(s_u - s_d)} = \frac{50 \cdot 150 - 100 \cdot 100}{1.02 \cdot 50} = -\frac{2500}{51} \approx -49.02.$$

No-arbitrage price of the claim at $t = 0$:

$$C_0 = \phi S_0 + \psi B_0 = 125 - 49.02 = 75.98.$$

Check (the portfolio reproduces the payoff in both states):

$$\phi s_u + \psi B_1 = 150 - 50 = 100 = c_u, \qquad \phi s_d + \psi B_1 = 100 - 50 = 50 = c_d.$$

Worked example: risk-neutral pricing

Risk-neutral probability (makes S/B a $\mathbb{Q}$-martingale):

$$q = \frac{\frac{B_1}{B_0} - \frac{s_d}{S_0}}{\frac{s_u}{S_0} - \frac{s_d}{S_0}} = \frac{1.02 - 0.80}{1.20 - 0.80} = \frac{0.22}{0.40} = 0.55.$$

Price as discounted $\mathbb{Q}$-expectation of the payoff:

$$C_0 = \frac{1}{B_1}\mathbb{E}_{\mathbb{Q}}[C_1] = \frac{0.55 \cdot 100 + 0.45 \cdot 50}{1.02} = \frac{77.5}{1.02} = 75.98,$$

identical to the replication price — as it must be.

Worked example: why a *different* measure?

The price is *not* the discounted real-world expectation. Under $\mathbb{P}$ ($p = 0.7$):

$$\frac{1}{B_1}\mathbb{E}_{\mathbb{P}}[S_1] = \frac{0.7 \cdot 150 + 0.3 \cdot 100}{1.02} = \frac{135}{1.02} = 132.35 \neq 125 = S_0,$$

$$\frac{1}{B_1}\mathbb{E}_{\mathbb{P}}[C_1] = \frac{0.7 \cdot 100 + 0.3 \cdot 50}{1.02} = \frac{85}{1.02} = 83.33 \neq 75.98.$$

Under the risk-neutral $\mathbb{Q}$ ($q = 0.55$):

$$\frac{1}{B_1}\mathbb{E}_{\mathbb{Q}}[S_1] = \frac{0.55 \cdot 150 + 0.45 \cdot 100}{1.02} = \frac{127.5}{1.02} = 125 = S_0.$$

Only $\mathbb{Q}$ reproduces today's traded prices; this is exactly why we price under $\mathbb{Q}$ rather than $\mathbb{P}$.

Quiz

Take the one-period model with $S_0 = 100$, $B_0 = 1$, $B_1 = 1$ (so $r = 0$), and

$$S_{1,u} = 120, \qquad S_{1,d} = 90.$$

1. Is the market free of arbitrage?
2. Find the risk-neutral probability q of the "up" state.
3. Price a call with strike $K = 100$ (payoffs $c_u = 20$, $c_d = 0$).

Method 2: risk-neutral pricing

First fundamental theorem of asset pricing (for the toy model)
The market is free of arbitrage opportunities if and only if there exists a probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ such that A/B is a martingale, under $\mathbb{Q}$, for all assets A.

Application We can price options by applying this theorem twice:

- Write down model for B and S (without arbitrage opportunities).
- Apply theorem (with $A = B$ and $A = S$; note that for $A = B$ the 'martingale-property' is trivial). This yields unique $\mathbb{Q}$. $(\star)$
- Now introduce an option to market with payoff C_1. We want to set price such that we do not create arbitrage opportunity.
- Apply theorem again: A/B should be a martingale for $A = B, S, C$. For $A = S$ we obtain unique $\mathbb{Q}$ from previous step $(\star)$. And $A = C$ yields

$$C_0 = B_0 \mathbb{E}_{\mathbb{Q}}[C_1/B_1].$$

Proof

'$\Rightarrow$': exercise - check that we have proved this implication (via explicit construction of $\mathbb{Q}$)

'$\Leftarrow$':

Consider portfolio V with $V_1 \geq 0$ and $V_1(\omega) > 0$ for at least one state-of-the-world. If we prove that $V_0 > 0$, then we have shown that arbitrage opportunities do not exist.

Note that we have (why?!)

$$
\begin{aligned}
0 < \mathbb{E}_{\mathbb{Q}}\left[\frac{V_1}{B_1}\right] &= \phi \mathbb{E}_{\mathbb{Q}}\left[\frac{S_1}{B_1}\right] + \psi \\
&= \phi \frac{S_0}{B_0} + \psi = \frac{1}{B_0}(\phi S_0 + \psi B_0) = \frac{V_0}{B_0},
\end{aligned}
$$

which yields $V_0 > 0$.

Method 3: the pricing kernel

A third, equivalent route. A **pricing kernel** (a.k.a. stochastic discount factor) is a random variable $m > 0$ such that, for *every* asset A,

$$A_0 = \mathbb{E}_{\mathbb{P}}[\, m\, A_1\,].$$

In the toy model m takes two values m_u, m_d, determined from the basic assets:

$$\begin{aligned} B_0 &= \mathbb{E}_{\mathbb{P}}[mB_1] = B_1\big(p\, m_u + (1-p)m_d\big), \\ S_0 &= \mathbb{E}_{\mathbb{P}}[mS_1] = p\, m_u s_u + (1-p)m_d s_d. \end{aligned}$$

Two equations, two unknowns. Then *any* claim is priced by

$$C_0 = \mathbb{E}_{\mathbb{P}}[m\, C_1] = p\, m_u c_u + (1-p)m_d c_d.$$

Method 3: relation to $\mathbb{Q}$, and the worked example

The three methods are one and the same. Solving the two equations gives

$$m_\omega = \frac{B_0}{B_1}\,\frac{q_\omega}{p_\omega}, \qquad \text{i.e.} \qquad m = \underbrace{\frac{B_0}{B_1}}_{\text{discount}} \times \underbrace{\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}}_{\text{change of measure}}.$$

For our numbers ($p = 0.7,\ q = 0.55,\ B_1 = 1.02$):

$$m_u = \frac{0.55}{1.02 \cdot 0.7} \approx 0.770, \qquad m_d = \frac{0.45}{1.02 \cdot 0.3} \approx 1.471,$$

$$C_0 = 0.7 \cdot 0.770 \cdot 100 + 0.3 \cdot 1.471 \cdot 50 \approx 75.98,$$

again the same price. Note $m_d > m_u$: a payoff in the "down" (low-S) state is more expensive — the kernel encodes the price of risk.

Section 3

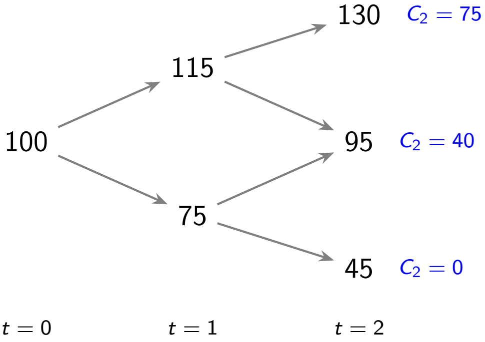
Multi-period model

Multi-period binomial model

The one-period idea extends directly to several trading dates. Consider a *two-period*, recombining model ($t = 0, 1, 2$): we may trade and rebalance at $t = 0$ and $t = 1$.

Bond: $B_0 = 1$, $B_1 = 1.05$, $B_2 = 1.11$.

Claim: call with $K = 55$, so the payoff $C_2 = \max\{S_2 - 55, 0\}$ equals $75, 40, 0$ in the three terminal states.



Pricing by backward induction

Key observation: each *one-period sub-tree* is exactly the model we already solved.

- start at maturity: the values at $t = 2$ are the known payoffs;
- step back one period at a time. At a node at time t with one-period successors "u" / "d", the no-arbitrage value is

$$V_t = \frac{B_t}{B_{t+1}}\left(q\, V^u_{t+1} + (1-q)\, V^d_{t+1}\right), \qquad q = \frac{B_{t+1}/B_t - s_d/S_t}{s_u/S_t - s_d/S_t},$$

  equivalently, replicate that single period with (ϕ, ψ);
- the resulting strategy is *self-financing*: at $t = 1$ we rebalance, and the value of the old position exactly funds the new one (no cash injected/withdrawn).

We illustrate with replication (the spreadsheet also verifies the risk-neutral route).

Two-period example: step back to $t = 1$

At each $t = 1$ node we solve a one-period problem for the $t = 2$ payoffs.

Up node ($S_{1,u} = 115 \to 130/95$, payoffs $75/40$):

$$\phi_{1,u} = \frac{75 - 40}{130 - 95} = 1, \qquad \psi_{1,u} \approx -49.55, \qquad V_{1,u} \approx 62.97.$$

Down node ($S_{1,d} = 75 \to 95/45$, payoffs $40/0$):

$$\phi_{1,d} = \frac{40 - 0}{95 - 45} = 0.8, \qquad \psi_{1,d} \approx -32.43, \qquad V_{1,d} \approx 25.95.$$

Two-period example: step back to $t = 0$

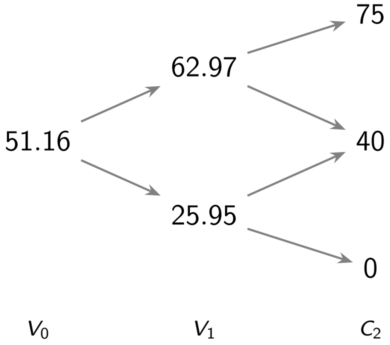
Now treat $V_{1,u} \approx 62.97$ and $V_{1,d} \approx 25.95$ as the "payoffs" of a one-period problem at $t = 0$:

$$\phi_0 = \frac{62.97 - 25.95}{115 - 75} \approx 0.926, \qquad \psi_0 \approx -41.41,$$

$$V_0 = \phi_0 S_0 + \psi_0 B_0 = 0.926 \cdot 100 - 41.41 \approx 51.16.$$

So the no-arbitrage price of the two-period call is ≈ 51.16.

At $t = 1$ we *rebalance* from (ϕ_0, ψ_0) to $(\phi_{1,\cdot}, \psi_{1,\cdot})$; self-financing means the rebalancing costs nothing.



Food for thought

Food for thought

Suppose the stock can move to *three* values at $t = 1$ ("up", "middle", "down") instead of two, with the bond unchanged.

- Can you still replicate an arbitrary payoff (c_u, c_m, c_d) using only the stock and the bond?
- How many probability measures $\mathbb{Q}$ make S/B a martingale now?
- What does this imply for the *uniqueness* of the no-arbitrage price?

This is the essence of *market incompleteness*: with more states than independent traded assets, replication fails and $\mathbb{Q}$ is no longer unique. We return to completeness in Part 3.

Toy model - concluding remarks

For the toy model we have seen that we can price options by

- constructing a replicating portfolio,
- using risk-neutral pricing, and
- using the pricing kernel.

All three give the *same* price, and they extend to the multi-period tree via backward induction.

For the continuous-time case we will see that exactly the same ideas are used. Unfortunately (?), the math gets a bit more difficult.

Section 4

Appendix

Appendix: proof of the no-arbitrage theorem

Claim. The market is free of arbitrage $\iff$ $0 < \dfrac{s_d}{S_0} < \dfrac{B_1}{B_0} < \dfrac{s_u}{S_0}$.

'$\Rightarrow$' (contrapositive). Suppose the chain of inequalities fails; recall $0 < s_d < s_u$.

- If $\dfrac{B_1}{B_0} \le \dfrac{s_d}{S_0}$: buy one share, financed by selling S_0/B_0 units of the bond, so $V_0 = 0$. Then
$$V_1 = S_1 - \tfrac{S_0}{B_0} B_1 \ \ge\ s_d - \tfrac{S_0}{B_0} B_1 \ \ge\ 0,$$
with strict inequality in the "up" state $\Rightarrow$ arbitrage.
- If $\dfrac{B_1}{B_0} \ge \dfrac{s_u}{S_0}$: the reverse strategy (short one share, invest S_0 in the bond) gives $V_0 = 0$ and $V_1 = \frac{S_0}{B_0} B_1 - S_1 \ge 0$, strict in the "down" state $\Rightarrow$ arbitrage.

So absence of arbitrage forces the strict inequalities.

'$\Leftarrow$' The inequalities are equivalent to $q \in (0,1)$; the risk-neutral construction (Method 2) then assigns every claim a price consistent with S and B, leaving no arbitrage. $\square$